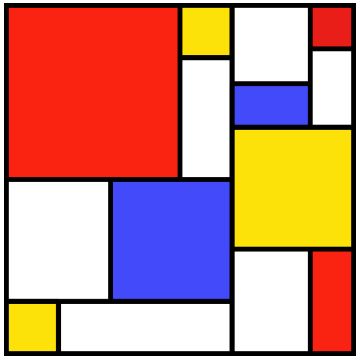
SLiM

Workshop Series



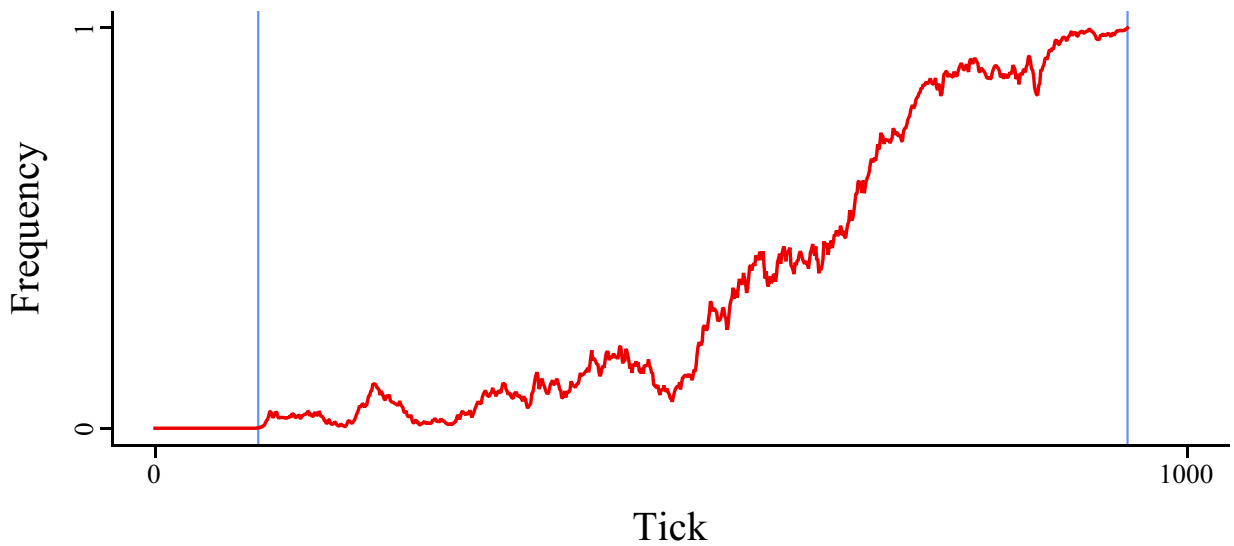
Selective Sweeps

Benjamin C. Haller

Messer Lab, Cornell University

Selective Sweeps

- a selective sweep model involves:
  - a **sweep mutation**, which is beneficial
  - followed until establishment / completion



Selective Sweeps

- two basic types of sweeps:
- **“hard”** sweeps
  - starting from a single ancestor
  - typically from a new mutation

Selective Sweeps

- two basic types of sweeps:
- **"hard"** sweeps
  - starting from a single ancestor
  - typically from a new mutation
- **"soft"** sweeps
  - starting from multiple ancestors
  - typically from standing genetic variation

Selective Sweeps

- to model a **hard sweep**:
  - choose **candidate targets**:
    - `p1.haplosomes`
    - `sim.subpopulations.haplosomes`

Selective Sweeps

- to model a **hard sweep**:
  - choose **candidate targets**:
    - `p1.haplosomes`
    - `sim.subpopulations.haplosomes`
  - choose the **target haplosome**:
    - `sample()`

Selective Sweeps

- to model a **hard sweep**:
  - choose **candidate targets**:
    - `p1.haplosomes`
    - `sim.subpopulations.haplosomes`
  - choose the **target haplosome**:
    - `sample()`
  - introduce the **sweep mutation**:
    - `addNewMutation()`
    - `addNewDrawnMutation()`

Selective Sweeps

- to model a **soft sweep**:
  - choose **candidate mutations**:
    - `sim.mutations`
    - `individual.haplosomes.mutations`

Selective Sweeps

- to model a **soft sweep**:
  - choose **candidate mutations**:
    - `sim.mutations`
    - `individual.haplosomes.mutations`
  - choose the **target mutation**:
    - `sample()`

Selective Sweeps

- to model a **soft sweep**:
  - choose **candidate mutations**:
    - `sim.mutations`
    - `individual.haplosomes.mutations`
  - choose the **target mutation**:
    - `sample()`
  - transmute the **sweep mutation**:
    - `setMutationType()`
    - `setSelectionCoefficient()`

Selective Sweeps

- you might want to know:
  - when it's lost / established / fixed

Selective Sweeps

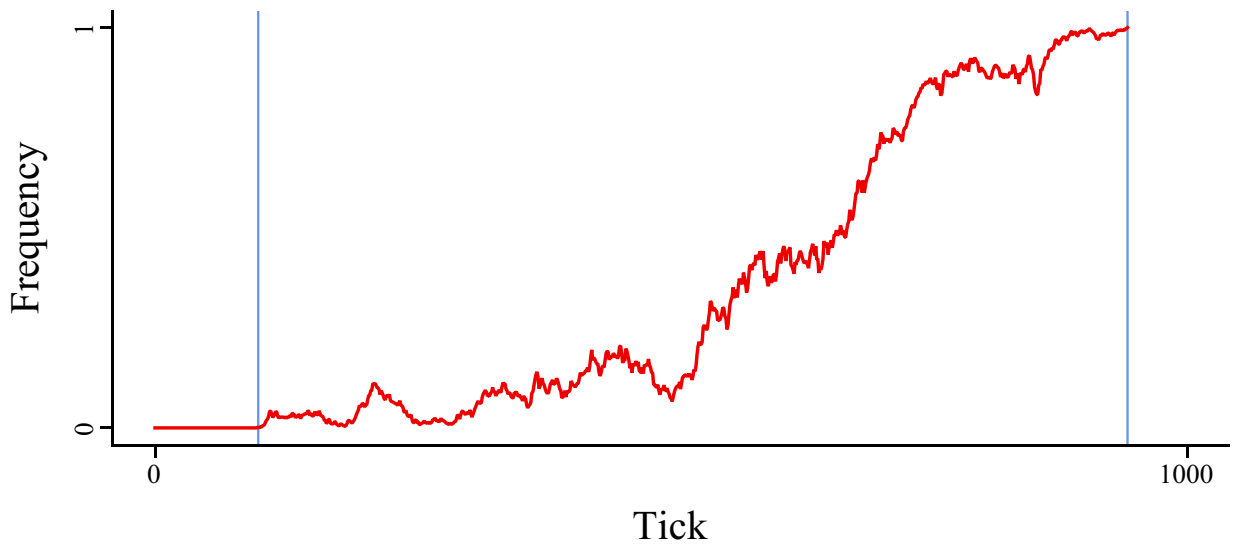
- you might want to know:
  - when it's lost / established / fixed
- you can monitor its progress:
  - `countOfMutationsOfType()` / `mutationsOfType()`
  - `mutationFrequencies()` / `mutationCounts()`

Selective Sweeps

- you might want to know:
  - when it's lost / established / fixed
- you can monitor its progress:
  - `countOfMutationsOfType()` / `mutationsOfType()`
  - `mutationFrequencies()` / `mutationCounts()`
- or keep a reference to the sweep mutation:
  - `defineConstant()` / `defineGlobal()`
  - `isSegregating` / `isFixed`

Conditional Simulations

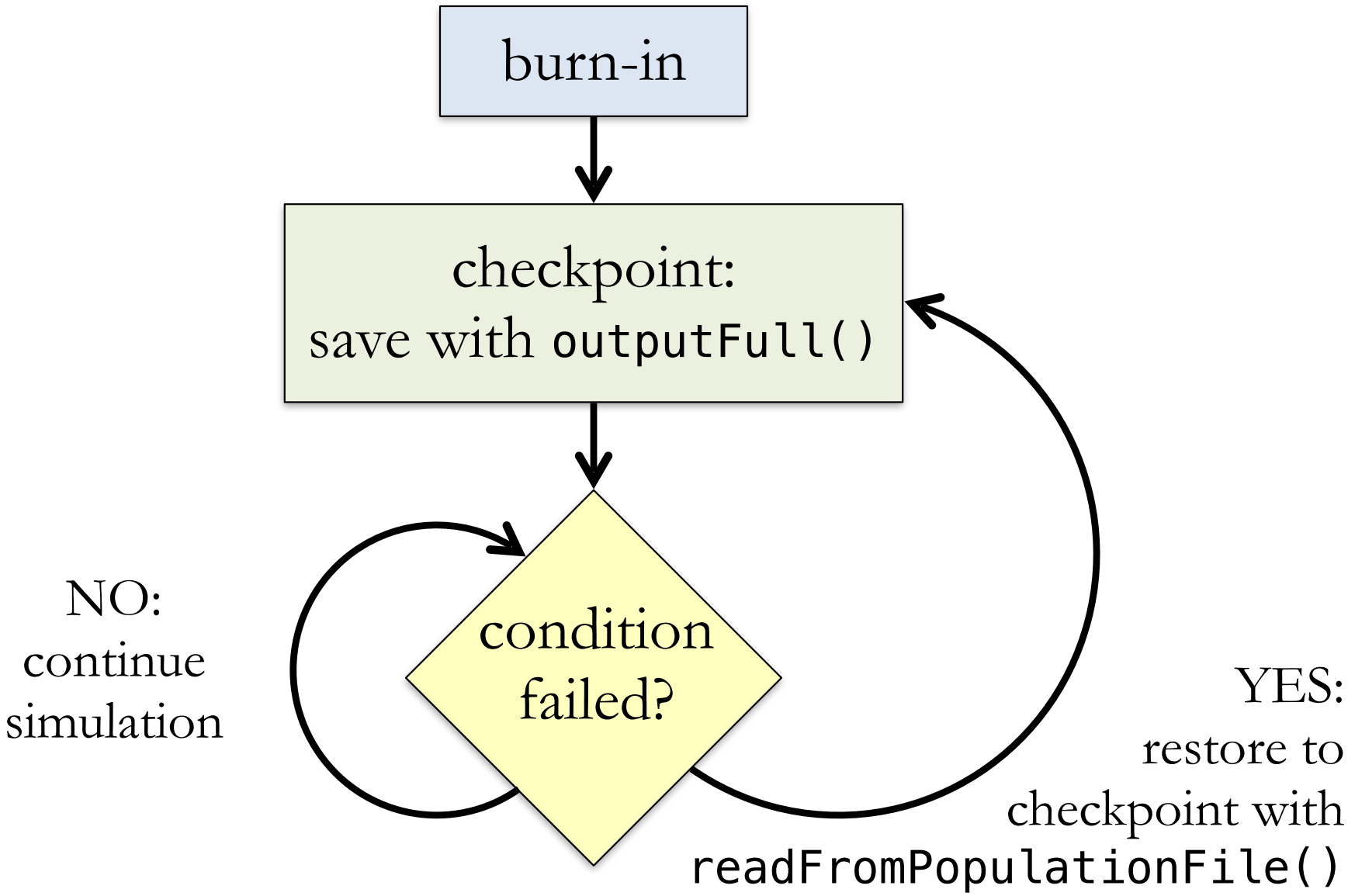
- often desirable to be "conditional"
  - guarantees a particular outcome
  - doesn't bias the process leading there



Conditional Simulations

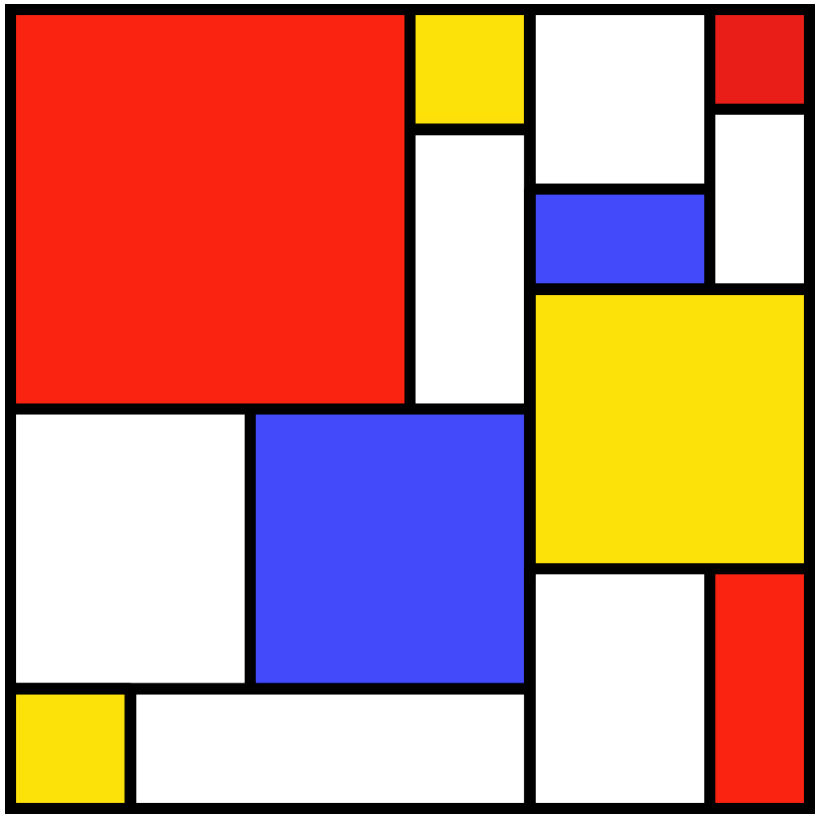
- how can we avoid biasing the process?
- **save** at a "checkpoint"
  - `outputFull()`
- **check** the condition periodically
- **reload** to the checkpoint as needed
  - `readFromPopulationFile()`

Conditional Simulations



Conditional Simulations

- we've been talking about sweeps, but…
- conditionality can be on **anything!**
  - whether extinction occurs
  - whether introgression establishes
  - whether a new habitat gets colonized
  - whether adaptation occurs fast enough



Copyright © 2020–2025 Benjamin C. Haller. All Rights Reserved.